

Metric criteria for distance-balanced graph variants^{*}

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Abstract

We address two problems raised in [4] and [5]. For quasi- λ -distance-balanced direct products, we prove a parity-based distance formula and obtain a factor-wise characterization, with explicit consequences for $K_{m,n} \times H$. For nicely distance-balanced graphs, we resolve the vertex-transitive case by showing that the property is equivalent to edge-independence of the equidistant-vertex number, yielding edge-orbit, local, Cayley graph and Cartesian-product criteria.

Keywords: distance-balanced graph, quasi- λ -distance-balanced graph, nicely distance-balanced graph, direct product, vertex-transitive graph, Cayley graph

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1. Introduction

Let Γ be a finite connected graph. For adjacent vertices $u, v \in V(\Gamma)$, set

$$W_{u,v}^\Gamma = \{z \in V(\Gamma) : d_\Gamma(z, u) < d_\Gamma(z, v)\}.$$

The graph Γ is *distance-balanced* if $|W_{u,v}^\Gamma| = |W_{v,u}^\Gamma|$ for every edge uv . Distance-balanced graphs were introduced and studied in several forms in [1, 2, 3]. Two refinements are central here. A graph is *quasi- λ -distance-balanced*, where $\lambda > 1$ is a positive rational number, if for every edge one of $|W_{u,v}|$ and $|W_{v,u}|$

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is λ times the other. A graph is *nicely distance-balanced* if there is a positive integer γ_Γ such that

$$|W_{u,v}^\Gamma| = |W_{v,u}^\Gamma| = \gamma_\Gamma$$

for every edge uv . These notions were introduced in [4] and [5], respectively.

We use the edge decomposition

$$V(\Gamma) = W_{u,v}^\Gamma \sqcup W_{v,u}^\Gamma \sqcup E_{u,v}^\Gamma, \quad E_{u,v}^\Gamma = \{z : d_\Gamma(z, u) = d_\Gamma(z, v)\},$$

and view the triple

$$(|W_{u,v}^\Gamma|, |W_{v,u}^\Gamma|, |E_{u,v}^\Gamma|)$$

as the metric profile of the edge. This language is elementary, but it separates two different sources of variation. In direct products, the main obstruction is the synchronization of walks in the two coordinates. In vertex-transitive graphs, the ordinary balance is automatic, and the remaining obstruction is the variation of the equidistance term $|E_{u,v}^\Gamma|$ over edge orbits.

Our first result answers the direct-product question posed in [4]. The direct product $G \times H$ has vertex set $V(G) \times V(H)$, with (g, h) adjacent to (g', h') if and only if $gg' \in E(G)$ and $hh' \in E(H)$. A walk of length k in $G \times H$ is a pair of walks of the same length k in the two factors, so parity has to be recorded. If $\delta_\Gamma^0(a, b)$ and $\delta_\Gamma^1(a, b)$ denote the lengths of the shortest even and odd a - b walks in a graph Γ , with value ∞ when no such walk exists, then

$$d_{G \times H}((g, h), (x, y)) = \min_{\varepsilon \in \{0,1\}} \max\{\delta_G^\varepsilon(g, x), \delta_H^\varepsilon(h, y)\}.$$

Substitution into the definition of $W_{u,v}$ gives a necessary and sufficient factor-wise criterion for $G \times H$ to be quasi- λ -distance-balanced. The criterion becomes explicit when one factor is complete bipartite: for a connected non-bipartite graph H and an oriented edge xy of H , put

$$\begin{aligned} \alpha_H(x, y) &= |\{z : \delta_H^0(x, z) < \delta_H^1(y, z)\}|, \\ \beta_H(x, y) &= |\{z : \delta_H^1(x, z) < \delta_H^0(y, z)\}|. \end{aligned}$$

Then $K_{m,n} \times H$ is controlled by

$$S(x, y) = m\alpha_H(x, y) + n\beta_H(x, y) + n - m.$$

This gives concrete families such as $K_{m,n} \times C_{2r+1}$ and $K_{m,n} \times K_q$, and a one-parameter reduction when H is arc-transitive.

The second part concerns the problem of characterizing nicely distance-balanced vertex-transitive graphs [5, Problem 6.2]. A finite connected vertex-transitive graph is strongly distance-balanced, and hence distance-balanced. Consequently, if Γ has order n and uv is an edge, then

$$n = |W_{u,v}^\Gamma| + |W_{v,u}^\Gamma| + |E_{u,v}^\Gamma| = 2|W_{u,v}^\Gamma| + |E_{u,v}^\Gamma|.$$

It follows that a finite connected vertex-transitive graph Γ is nicely distance-balanced if and only if $|E_{u,v}^\Gamma|$ is independent of the edge uv . If the common equidistance value is η , then

$$\gamma_\Gamma = \frac{|V(\Gamma)| - \eta}{2}.$$

This converts the vertex-transitive problem into an edge-orbit invariant. We derive a local test using a vertex stabilizer, a word-length criterion for Cayley graphs, and a Cartesian-product formulation. We also give infinite families of non-bipartite vertex-transitive NDB graphs which are not edge-transitive; thus the vertex- and edge-transitive sufficient condition from [5, Proposition 6.3] is not necessary.

The paper is organized as follows. Section 2 introduces edge profiles and the balance variants. Section 3 proves the direct-product distance formula and the quasi- λ criterion. Section 4 specializes this criterion to complete bipartite factors and arc-transitive factors. Section 5 proves the equidistance-set criterion for vertex-transitive NDB graphs. Section 6 gives the Cayley-graph formulation and treats cubic circulants. Section 7 records product constructions of vertex-transitive NDB graphs that are not edge-transitive. Section 8 recalls the relation with the non-SDB problem of Kutnar and Miklavíč, which has been solved by Fernández and Hujdurović [6].

2. Metric profiles and balance variants

All graphs in the paper are finite, simple and undirected. Unless stated otherwise, they are connected and have at least one edge. Distances in a graph Γ are denoted by d_Γ , or simply by d when the graph is clear. The order of a graph is its number of vertices.

Definition 2.1. *Let uv be an edge of a connected graph Γ . The oriented metric profile of the ordered edge (u, v) is*

$$\pi_\Gamma(u, v) = (w_\Gamma(u, v), w_\Gamma(v, u), \eta_\Gamma(u, v)),$$

where

$$w_\Gamma(u, v) = |W_{u,v}^\Gamma|, \quad \eta_\Gamma(u, v) = |E_{u,v}^\Gamma|,$$

and

$$E_{u,v}^\Gamma = \{z \in V(\Gamma) : d_\Gamma(z, u) = d_\Gamma(z, v)\}.$$

The unordered edge profile is obtained by identifying (a, b, c) with (b, a, c) .

For every edge uv of Γ ,

$$V(\Gamma) = W_{u,v}^\Gamma \sqcup W_{v,u}^\Gamma \sqcup E_{u,v}^\Gamma. \quad (1)$$

Thus

$$w_\Gamma(u, v) + w_\Gamma(v, u) + \eta_\Gamma(u, v) = |V(\Gamma)|. \quad (2)$$

Definition 2.2. A connected graph Γ is

- (i) distance-balanced if $w_\Gamma(u, v) = w_\Gamma(v, u)$ for every edge uv ;
- (ii) quasi- λ -distance-balanced, for a rational number $\lambda > 1$, if for every edge uv ,

$$\frac{\max\{w_\Gamma(u, v), w_\Gamma(v, u)\}}{\min\{w_\Gamma(u, v), w_\Gamma(v, u)\}} = \lambda;$$

- (iii) nicely distance-balanced, abbreviated NDB, if there exists an integer $\gamma_\Gamma > 0$ such that

$$w_\Gamma(u, v) = w_\Gamma(v, u) = \gamma_\Gamma$$

for every edge uv .

The following observation is the profile form of these definitions.

Proposition 2.3. Let Γ be a connected graph of order N .

- (a) Γ is distance-balanced if and only if every edge profile has the form $(a, a, N - 2a)$.
- (b) Γ is NDB if and only if every edge profile has the same form $(\gamma_\Gamma, \gamma_\Gamma, N - 2\gamma_\Gamma)$.
- (c) If Γ is distance-balanced, then Γ is NDB if and only if $\eta_\Gamma(u, v)$ is independent of the edge uv .

Proof. Parts (a) and (b) are restatements of the definitions. For (c), if Γ is distance-balanced, then (2) gives

$$w_\Gamma(u, v) = w_\Gamma(v, u) = \frac{N - \eta_\Gamma(u, v)}{2}.$$

Thus the common value is independent of uv if and only if $\eta_\Gamma(u, v)$ is independent of uv . $\square$

We shall also use the strongly distance-balanced condition. For vertices u, v of Γ and integers $i, j \geq 0$, put

$$D_{i,j}^\Gamma(u, v) = \{x \in V(\Gamma) : d_\Gamma(x, u) = i, d_\Gamma(x, v) = j\}.$$

A connected graph Γ is *strongly distance-balanced*, abbreviated *SDB*, if for every edge uv and every $i \geq 0$,

$$|D_{i,i+1}^\Gamma(u, v)| = |D_{i+1,i}^\Gamma(u, v)|.$$

Every SDB graph is distance-balanced, by summing over i .

For direct products we need shortest walks with prescribed parity. If Γ is a connected non-trivial graph, $a, b \in V(\Gamma)$, and $\varepsilon \in \{0, 1\}$, define

$$\delta_\Gamma^\varepsilon(a, b) = \min\{\ell : \text{there is an } a\text{-}b \text{ walk of length } \ell \text{ and } \ell \equiv \varepsilon \pmod{2}\}, \quad (3)$$

with value ∞ if no such walk exists. In a bipartite graph exactly one of $\delta_\Gamma^0(a, b)$ and $\delta_\Gamma^1(a, b)$ is finite. In a connected non-bipartite graph both are finite for every pair a, b .

We use the extended-distance convention that $\max\{\infty, t\} = \infty$ and that the minimum of two infinite quantities is ∞ .

3. Direct products and quasi-balance

The direct product $G \times H$ has vertex set $V(G) \times V(H)$, and

$$(g, h) \sim (g', h') \iff gg' \in E(G) \text{ and } hh' \in E(H).$$

The essential metric feature is that a product walk is a pair of coordinate walks of the same length.

Lemma 3.1 (Direct-product distance formula). *Let G and H be connected non-trivial graphs. For all $(g, h), (x, y) \in V(G \times H)$,*

$$d_{G \times H}((g, h), (x, y)) = \min_{\varepsilon \in \{0,1\}} \max\{\delta_G^\varepsilon(g, x), \delta_H^\varepsilon(h, y)\}, \quad (4)$$

where the left-hand side is understood as ∞ if the two product vertices are in different connected components.

Proof. A walk of length k from (g, h) to (x, y) in $G \times H$ is precisely a pair consisting of a g - x walk of length k in G and an h - y walk of length k in H . Thus k has a parity ε for which both $\delta_G^\varepsilon(g, x)$ and $\delta_H^\varepsilon(h, y)$ are finite, and

$$k \geq \max\{\delta_G^\varepsilon(g, x), \delta_H^\varepsilon(h, y)\}.$$

This proves the lower bound and also shows that no product walk exists if both parity choices yield ∞ .

Conversely, fix ε for which both parity distances are finite and let

$$M = \max\{\delta_G^\varepsilon(g, x), \delta_H^\varepsilon(h, y)\}.$$

The two quantities inside the maximum have parity ε , so M has parity ε . Take shortest walks of parity ε in the two factors. If one walk is shorter than M , insert backtracking subwalks $aa'a$ as needed; each insertion increases the length by 2. Since the factors are connected and non-trivial, every vertex of such a walk has a neighbour and the insertion is available. We obtain coordinate walks of the same length M , hence a product walk of length M . Taking the minimum over ε proves (4). $\square$

For $p = (g, h) \in V(G \times H)$ and $(x, y) \in V(G) \times V(H)$, put

$$\Phi_p(x, y) = \min_{\varepsilon \in \{0,1\}} \max\{\delta_G^\varepsilon(g, x), \delta_H^\varepsilon(h, y)\}. \quad (5)$$

When the direct product is connected, $\Phi_p(x, y)$ is the distance from p to (x, y) .

For an edge

$$e = pq = ((g, h), (g', h')) \in E(G \times H),$$

define

$$A_e = |\{(x, y) \in V(G) \times V(H) : \Phi_p(x, y) < \Phi_q(x, y)\}|, \quad (6)$$

$$B_e = |\{(x, y) \in V(G) \times V(H) : \Phi_q(x, y) < \Phi_p(x, y)\}|. \quad (7)$$

Theorem 3.2 (Quasi-balance criterion for direct products). *Let G and H be connected non-trivial graphs, and let $\lambda > 1$ be rational. Then $G \times H$ is quasi- λ -distance-balanced if and only if the following conditions hold:*

- (i) *at least one of G and H is non-bipartite;*
- (ii) *for every edge $e \in E(G \times H)$, the integers A_e and B_e defined in (6) and (7) satisfy*

$$\frac{\max\{A_e, B_e\}}{\min\{A_e, B_e\}} = \lambda. \quad (8)$$

Equivalently, $G \times H$ is quasi- λ -distance-balanced if and only if it is connected and the ratio set

$$\left\{ \frac{\max\{A_e, B_e\}}{\min\{A_e, B_e\}} : e \in E(G \times H) \right\}$$

consists of the single value $\lambda > 1$.

Proof. The direct product of two connected non-trivial graphs is connected if and only if at least one factor is non-bipartite [7, Theorem 5.9]. Thus (i) is exactly the connectedness requirement in the definition of quasi- λ -distance-balance.

Assume $G \times H$ is connected and let $e = pq$ be an edge. By Lemma 3.1, Φ_p and Φ_q are the distance functions from p and q in $G \times H$. Hence

$$A_e = |W_{p,q}^{G \times H}|, \quad B_e = |W_{q,p}^{G \times H}|.$$

Both numbers are positive because $p \in W_{p,q}$ and $q \in W_{q,p}$. Condition (8) is therefore exactly the quasi- λ condition for the edge pq . Requiring it for all edges proves the theorem. $\square$

Remark 3.3. *Theorem 3.2 is a factor-wise metric criterion. Its input consists only of shortest even and odd walks in G and H . This is the precise point at which the direct product differs from the Cartesian product, whose distance is $d_G + d_H$.*

The criterion can be reduced by symmetry.

Proposition 3.4 (Orbit reduction). *Let G and H be connected non-trivial graphs. The pair $\{A_e, B_e\}$ is constant on the orbits of $\text{Aut}(G) \times \text{Aut}(H)$ on the edge set of $G \times H$. In particular, to test the condition in Theorem 3.2, it is enough to check one representative from each such orbit.*

Proof. Automorphisms of G and H preserve all parity distances δ^ε . Hence the map induced by $(\alpha, \beta) \in \text{Aut}(G) \times \text{Aut}(H)$ sends the inequalities defining A_e and B_e bijectively to the corresponding inequalities for the image edge. The unordered pair is preserved because an automorphism of the product may send the chosen orientation of an edge to either orientation of its image. $\square$

Abadi et al. proved that every quasi- λ -distance-balanced graph is triangle-free [4]. Since $G \times H$ contains a triangle if and only if both G and H contain a triangle, we have the following immediate obstruction.

Corollary 3.5. *If $G \times H$ is quasi- λ -distance-balanced, then at least one of G and H is triangle-free.*

4. A complete bipartite factor

We now make Theorem 3.2 explicit for products with a complete bipartite factor. Let $K_{m,n}$ have bipartition $X \cup Y$, with $|X| = m$ and $|Y| = n$. Throughout this section, H is a connected non-bipartite graph of order N ; this guarantees that $K_{m,n} \times H$ is connected.

For an oriented edge xy of H , define

$$\alpha_H(x, y) = |\{z \in V(H) : \delta_H^0(x, z) < \delta_H^1(y, z)\}|, \quad (9)$$

$$\beta_H(x, y) = |\{z \in V(H) : \delta_H^1(x, z) < \delta_H^0(y, z)\}|. \quad (10)$$

Theorem 4.1 (Complete bipartite factor). *Let $m, n \geq 1$, and let H be a connected non-bipartite graph of order N . For each oriented edge xy of H , put*

$$S(x, y) = m\alpha_H(x, y) + n\beta_H(x, y) + n - m. \quad (11)$$

Then $K_{m,n} \times H$ is quasi- λ -distance-balanced if and only if there is an integer s with

$$0 < s < \frac{(m+n)N}{2} \quad (12)$$

such that

$$S(x, y) \in \{s, (m+n)N - s\} \quad (13)$$

for every oriented edge xy of H . In that case

$$\lambda = \frac{(m+n)N - s}{s}. \quad (14)$$

Proof. Fix $x_0 \in X$, $y_0 \in Y$, and an oriented edge xy of H . Consider the product edge

$$e = ((x_0, x), (y_0, y)).$$

Every edge of $K_{m,n} \times H$ is of this form after choosing an orientation from the X -part to the Y -part in the $K_{m,n}$ coordinate and an orientation in H .

The parity distances from x_0 in $K_{m,n}$ are

$$\delta_{K_{m,n}}^0(x_0, u) = \begin{cases} 0, & u = x_0, \\ 2, & u \in X \setminus \{x_0\}, \\ \infty, & u \in Y, \end{cases} \quad \delta_{K_{m,n}}^1(x_0, u) = \begin{cases} 1, & u \in Y, \\ \infty, & u \in X. \end{cases}$$

Analogous formulas hold from y_0 , with X and Y interchanged.

Let $z \in V(H)$. First consider vertices (u, z) with $u \in X$. If $u = x_0$, Lemma 3.1 gives

$$d((x_0, x), (u, z)) = \delta_H^0(x, z), \quad d((y_0, y), (u, z)) = \delta_H^1(y, z),$$

so the contribution to $W_{(x_0, x), (y_0, y)}^{K_{m,n} \times H}$ is $\alpha_H(x, y)$. If $u \in X \setminus \{x_0\}$, then

$$d((x_0, x), (u, z)) = \max\{2, \delta_H^0(x, z)\}, \quad d((y_0, y), (u, z)) = \delta_H^1(y, z).$$

For $z \neq x$, the value $\delta_H^0(x, z)$ is at least 2, so this gives the same inequality as in (9). For $z = x$, the inequality $\delta_H^0(x, x) < \delta_H^1(y, x)$ holds, but

$$\max\{2, \delta_H^0(x, x)\} < \delta_H^1(y, x)$$

does not, since xy is an edge and $\delta_H^1(y, x) = 1$. Hence the total contribution from $X \times V(H)$ is

$$\alpha_H(x, y) + (m-1)(\alpha_H(x, y) - 1) = m\alpha_H(x, y) - m + 1. \quad (15)$$

Now consider vertices (u, z) with $u \in Y$. If $u = y_0$, the relevant inequality is

$$\delta_H^1(x, z) < \delta_H^0(y, z),$$

which contributes $\beta_H(x, y)$. If $u \in Y \setminus \{y_0\}$, then

$$d((x_0, x), (u, z)) = \delta_H^1(x, z), \quad d((y_0, y), (u, z)) = \max\{2, \delta_H^0(y, z)\}.$$

For $z \neq y$, the value $\delta_H^0(y, z)$ is at least 2, so this is the same inequality as in (10). For $z = y$, there is one additional solution, because

$$\delta_H^1(x, y) = 1 < 2 = \max\{2, \delta_H^0(y, y)\},$$

whereas $\delta_H^1(x, y) < \delta_H^0(y, y)$ is false. The contribution from $Y \times V(H)$ is therefore

$$\beta_H(x, y) + (n - 1)(\beta_H(x, y) + 1) = n\beta_H(x, y) + n - 1. \quad (16)$$

Adding (15) and (16) yields the expanded count

$$\begin{aligned} |W_{(x_0, x), (y_0, y)}^{K_{m, n} \times H}| &= (m\alpha_H(x, y) - m + 1) + (n\beta_H(x, y) + n - 1) \\ &= m\alpha_H(x, y) + n\beta_H(x, y) + n - m \\ &= S(x, y). \end{aligned} \quad (17)$$

The graph $K_{m, n} \times H$ is bipartite, with parts $X \times V(H)$ and $Y \times V(H)$. Hence no vertex is equidistant from the ends of an edge, and

$$|W_{(y_0, y), (x_0, x)}^{K_{m, n} \times H}| = (m + n)N - S(x, y).$$

The quasi- λ condition is therefore equivalent to saying that the unordered pair

$$\{S(x, y), (m + n)N - S(x, y)\}$$

has the same ratio for every oriented edge xy of H . This is exactly (13), and (14) gives the common ratio. The strict bound (12) is the requirement $\lambda > 1$ rather than the distance-balanced case. $\square$

The quantities α_H and β_H simplify for symmetric factors. Recall that an arc is an ordered edge.

Lemma 4.2. *Let H be a connected non-bipartite vertex-transitive graph of order N . Then for every arc xy of H ,*

$$\alpha_H(x, y) + \beta_H(x, y) = N. \quad (18)$$

Proof. The graph $K_2 \times H$ is the canonical bipartite double cover of H . Since H is non-bipartite, this product is connected. Since H is vertex-transitive, $K_2 \times H$ is vertex-transitive as well: automorphisms of H act on the second coordinate and the map that swaps the two K_2 -layers is an automorphism.

As a finite connected vertex-transitive graph, $K_2 \times H$ is distance-balanced; this also follows from the ball argument in Lemma 5.1. It is bipartite, so for every edge no vertex can be equidistant from its two endpoints: the two distances have opposite parity. Thus, for an edge pq of $K_2 \times H$, the sets $W_{p,q}$ and $W_{q,p}$ form a partition of the $2N$ vertices and have equal cardinality. Each therefore has cardinality N .

Apply (17) with $m = n = 1$ to the edge $((x_0, x), (y_0, y))$ of $K_2 \times H$. In this case

$$|W_{(x_0, x), (y_0, y)}^{K_2 \times H}| = \alpha_H(x, y) + \beta_H(x, y),$$

and the preceding paragraph shows that this number is N . $\square$

Theorem 4.3 (Arc-transitive factors). *Let H be a connected non-bipartite arc-transitive graph of order N . Let*

$$\alpha = \alpha_H(x, y)$$

for any arc xy of H . Then $K_{m,n} \times H$ is distance-balanced if

$$(m - n) \left(\alpha - \frac{N}{2} - 1 \right) = 0, \quad (19)$$

and otherwise it is quasi- λ -distance-balanced, where

$$\lambda = \frac{\frac{(m+n)N}{2} + \left| (m-n) \left(\alpha - \frac{N}{2} - 1 \right) \right|}{\frac{(m+n)N}{2} - \left| (m-n) \left(\alpha - \frac{N}{2} - 1 \right) \right|}. \quad (20)$$

In particular, if N is odd, then $K_{m,n} \times H$ is quasi- λ -distance-balanced for every $m \neq n$.

Proof. Since H is arc-transitive, $\alpha_H(x, y)$ is independent of the arc xy . By Lemma 4.2, $\beta_H(x, y) = N - \alpha$ for every arc. Thus

$$S(x, y) = m\alpha + n(N - \alpha) + n - m$$

is independent of the arc, and Theorem 4.1 applies. The deviation from half the order is

$$S(x, y) - \frac{(m+n)N}{2} = (m-n) \left(\alpha - \frac{N}{2} - 1 \right).$$

If this is zero, the product is distance-balanced. If it is nonzero, the two W -set sizes are the two complementary values around $(m+n)N/2$, giving (20). If N is odd, the number $N/2 + 1$ is not an integer, whereas α is an integer, so (19) holds only when $m = n$. $\square$

4.1. Odd cycles and complete graphs

Corollary 4.4 (Odd cycles). *Let $r \geq 1$ and $m, n \geq 1$. The direct product $K_{m,n} \times C_{2r+1}$ is quasi- λ -distance-balanced if and only if $m \neq n$. If $M = \max\{m, n\}$ and $L = \min\{m, n\}$, then*

$$\lambda = \frac{(r+1)M + rL}{rM + (r+1)L}. \quad (21)$$

If $m = n$, the product is distance-balanced.

Proof. Let xy be an arc of C_{2r+1} . Deleting the edge xy leaves a path

$$x = p_0, p_1, \dots, p_{2r} = y.$$

For $z = p_i$, the two x - z arcs have lengths i and $2r+1-i$, while the two y - z arcs have lengths $2r-i$ and $i+1$. It follows that $\delta^0(x, z) < \delta^1(y, z)$ holds for exactly the $r+1$ even values of i , so $\alpha_{C_{2r+1}}(x, y) = r+1$. Similarly, $\delta^1(x, z) < \delta^0(y, z)$ holds for exactly the r odd values of i , so $\beta_{C_{2r+1}}(x, y) = r$. Theorem 4.1 gives

$$S(x, y) = m(r+1) + nr + n - m = rm + (r+1)n.$$

The complementary value is $(r+1)m + rn$. These are equal precisely when $m = n$, and their ratio is (21). $\square$

Corollary 4.5 (Complete graphs). *Let $q \geq 3$ and $m, n \geq 1$. The direct product $K_{m,n} \times K_q$ is quasi- λ -distance-balanced if and only if $m \neq n$. If $M = \max\{m, n\}$ and $L = \min\{m, n\}$, then*

$$\lambda = \frac{L + (q-1)M}{(q-1)L + M}. \quad (22)$$

If $m = n$, the product is distance-balanced.

Proof. For an arc xy of K_q , the inequality $\delta^0(x, z) < \delta^1(y, z)$ holds exactly for $z = x$ and $z = y$, so $\alpha_{K_q}(x, y) = 2$. The inequality $\delta^1(x, z) < \delta^0(y, z)$ holds exactly for the $q-2$ vertices $z \notin \{x, y\}$, so $\beta_{K_q}(x, y) = q-2$. Therefore

$$S(x, y) = 2m + n(q-2) + n - m = m + (q-1)n.$$

The complementary value is $(q-1)m + n$. The two values differ precisely when $m \neq n$, and the ratio is (22). $\square$

Example 4.6. *Corollaries 4.4 and 4.5 give*

$$K_{1,2} \times C_5 \text{ is quasi-}\frac{8}{7}\text{-distance-balanced,}$$

$$K_{1,2} \times K_3 \text{ is quasi-}\frac{5}{4}\text{-distance-balanced.}$$

These examples show that quasi- λ -balance of a direct product need not be inherited from quasi- λ -balance of the factors.

5. Vertex-transitive NDB graphs

We now turn from direct products to vertex-transitive graphs. The key point is that vertex-transitivity already supplies the ordinary balance.

Lemma 5.1. *Every finite connected vertex-transitive graph is SDB.*

Proof. Let uv be an edge of Γ , and let

$$B_i(x) = \{z \in V(\Gamma) : d(z, x) \leq i\}$$

be the ball of radius i around x . Vertex-transitivity gives $|B_i(u)| = |B_i(v)|$ for every $i \geq 0$. Since the two balls have the same size,

$$|B_i(u) \setminus B_i(v)| = |B_i(v) \setminus B_i(u)|.$$

We prove that

$$B_i(u) \setminus B_i(v) = D_{i,i+1}(u, v).$$

Let $z \in B_i(u) \setminus B_i(v)$. Then $d(z, u) \leq i$ and $d(z, v) > i$. Since u and v are adjacent, the triangle inequality gives

$$d(z, v) \leq d(z, u) + 1 \leq i + 1.$$

Combining $d(z, v) > i$ with $d(z, v) \leq i + 1$, and using integrality of distances, we get $d(z, v) = i + 1$. It remains to see that $d(z, u) = i$. If $d(z, u) < i$, then $d(z, u) \leq i - 1$, and the triangle inequality would give

$$d(z, v) \leq d(z, u) + 1 \leq i,$$

contradicting $d(z, v) > i$. Hence $d(z, u) = i$, so $z \in D_{i,i+1}(u, v)$.

Conversely, if $z \in D_{i,i+1}(u, v)$, then $d(z, u) = i \leq i$ and $d(z, v) = i + 1 > i$, so $z \in B_i(u) \setminus B_i(v)$. Thus $B_i(u) \setminus B_i(v) = D_{i,i+1}(u, v)$. The same argument with u and v interchanged gives

$$B_i(v) \setminus B_i(u) = D_{i+1,i}(u, v).$$

Therefore

$$|D_{i,i+1}(u, v)| = |D_{i+1,i}(u, v)|$$

for every $i \geq 0$, as required. $\square$

Theorem 5.2 (Equidistance-set criterion). *Let Γ be a finite connected vertex-transitive graph of order n . Then Γ is NDB if and only if $\eta_\Gamma(u, v)$ is independent of the edge uv . If the common value is η , then*

$$\gamma_\Gamma = \frac{n - \eta}{2}. \quad (23)$$

Proof. By Lemma 5.1, Γ is distance-balanced. Hence for every edge uv ,

$$|W_{u,v}^\Gamma| = |W_{v,u}^\Gamma|.$$

Using the disjoint decomposition (1), we obtain

$$n = |W_{u,v}^\Gamma| + |W_{v,u}^\Gamma| + \eta_\Gamma(u, v) = 2|W_{u,v}^\Gamma| + \eta_\Gamma(u, v).$$

Thus

$$|W_{u,v}^\Gamma| = |W_{v,u}^\Gamma| = \frac{n - \eta_\Gamma(u, v)}{2}.$$

The common value is independent of uv if and only if $\eta_\Gamma(u, v)$ is independent of uv , which is exactly the NDB condition. Formula (23) follows. $\square$

Corollary 5.3 (Edge-orbit criterion). *Let Γ be a finite connected vertex-transitive graph, and let*

$$e_1 = u_1v_1, \dots, e_t = u_tv_t$$

be representatives of the orbits of $\text{Aut}(\Gamma)$ on unordered edges. Then Γ is NDB if and only if

$$\eta_\Gamma(u_1, v_1) = \eta_\Gamma(u_2, v_2) = \dots = \eta_\Gamma(u_t, v_t).$$

Proof. For every automorphism $\alpha \in \text{Aut}(\Gamma)$,

$$\alpha(E_{u,v}^\Gamma) = E_{\alpha(u),\alpha(v)}^\Gamma.$$

Thus $\eta_\Gamma(u, v)$ is constant on each edge orbit. The result follows from Theorem 5.2. $\square$

Corollary 5.4 (Local form). *Let Γ be a finite connected vertex-transitive graph, let $o \in V(\Gamma)$, and let $A = \text{Aut}(\Gamma)_o$ be the stabilizer of o . Let*

$$a_1, \dots, a_t$$

be representatives of the orbits of A on the neighbourhood $\Gamma(o)$. Then Γ is NDB if and only if

$$|\{x \in V(\Gamma) : d(x, o) = d(x, a_i)\}|$$

is independent of $i = 1, \dots, t$.

Proof. Every edge can be mapped by an automorphism to an edge incident with o . Once one endpoint is fixed to be o , the remaining freedom is exactly the action of $\text{Aut}(\Gamma)_o$ on the neighbours of o . Thus the pairs oa_i represent the edge orbits, and the edge-orbit criterion applies. $\square$

Corollary 5.5. *Every finite connected bipartite vertex-transitive graph of order n is NDB, with*

$$\gamma_\Gamma = \frac{n}{2}.$$

Proof. If uv is an edge in a bipartite graph, then no vertex can be equidistant from u and v , because the two distances have opposite parity. Thus $\eta_\Gamma(u, v) = 0$ for every edge. Theorem 5.2 gives the result. $\square$

Remark 5.6 (Algorithmic form). *For a vertex-transitive graph with known edge-orbit representatives, the edge-orbit criterion gives a direct test: run breadth-first search from the two endpoints of each representative edge and count equidistant vertices. If Γ has N vertices, M edges, and t edge orbits, this takes $O(t(N + M))$ time after the edge-orbit representatives have been found. Thus the metric part of the NDB test is small for vertex-transitive graphs with few edge orbits.*

6. Cayley graphs and circulants

Let G be a finite group and let $S \subseteq G$ satisfy $S = S^{-1}$, $1 \notin S$, and $\langle S \rangle = G$. We use the right Cayley graph convention: $\text{Cay}(G, S)$ has vertex set G and edges $g \sim gs$ for $g \in G$ and $s \in S$. Let $\ell_S(g)$ be the word length of g with respect to S .

Corollary 6.1 (Cayley graph criterion). *Let $\Gamma = \text{Cay}(G, S)$ be a connected undirected Cayley graph. For $s \in S$, put*

$$\eta_S(s) = |\{g \in G : \ell_S(g) = \ell_S(s^{-1}g)\}|. \quad (24)$$

Then Γ is NDB if and only if $\eta_S(s)$ is independent of $s \in S$. In that case

$$\gamma_\Gamma = \frac{|G| - \eta_S(s)}{2}.$$

Proof. The left regular action of G maps every edge of $\text{Cay}(G, S)$ to an edge of the form $\{1, s\}$, with $s \in S$. For a vertex $g \in G$,

$$d(g, 1) = \ell_S(g^{-1}) = \ell_S(g),$$

and

$$d(g, s) = \ell_S(g^{-1}s) = \ell_S((g^{-1}s)^{-1}) = \ell_S(s^{-1}g),$$

because $S = S^{-1}$. Hence the number of vertices equidistant from the endpoints of $\{1, s\}$ is exactly $\eta_S(s)$. Theorem 5.2 proves the claim. $\square$

Corollary 6.2. *Let $\text{Aut}(G, S)$ denote the group of automorphisms of G that preserve S setwise. If $\text{Aut}(G, S)$ is transitive on S , then $\text{Cay}(G, S)$ is NDB.*

Proof. Elements of $\text{Aut}(G, S)$ preserve word length, and hence preserve the values $\eta_S(s)$. If the action on S is transitive, all values in (24) are equal. $\square$

Example 6.3 (Odd prisms). *Let $n = 2q + 1$ be odd. The generalized Petersen graph $GP(n, 1)$ is the prism $C_n \square K_2$ and is vertex-transitive. It is not NDB. For a vertical edge joining the two copies of the same vertex of C_n , there is no equidistant vertex. For an edge lying in one of the two n -cycles, the unique vertex of C_n equidistant from the two endpoints gives one equidistant vertex in each layer, so the equidistance number is 2. Thus $\eta(u, v)$ is not constant on edges. This recovers [5, Lemma 3.1] through Theorem 5.2.*

We next recover the classification of cubic circulant NDB graphs from [5, Proposition 6.5] by using the Cayley criterion. Here $\text{Cir}(n, S)$ denotes $\text{Cay}(\mathbb{Z}_n, S)$, where all residues are taken modulo n , and $\gcd(r, s)$ denotes the greatest common divisor of the integers r and s . In particular, $\gcd(r, s) = 1$ means that r and s are relatively prime.

Theorem 6.4 (Cubic circulants). *Let $n \geq 4$ be even, and let*

$$\Gamma = \text{Cir}(n, \{\pm a, n/2\})$$

be a connected cubic circulant. Then Γ is NDB if and only if $\gcd(n, a) = 1$.

Proof. Suppose first that $\gcd(n, a) = 1$. Multiplication by a^{-1} in $\mathbb{Z}_n$ gives an isomorphism to $\text{Cir}(n, \{\pm 1, n/2\})$; indeed $a^{-1}n/2 \equiv n/2 \pmod{n}$ because a^{-1} is odd. If $n \equiv 2 \pmod{4}$, all three generators 1, -1 and $n/2$ are odd, so the graph is bipartite with bipartition given by parity. It is therefore NDB by the bipartite vertex-transitive corollary.

Assume now that $n \equiv 0 \pmod{4}$, and write $\rho = n/4$. In $\text{Cir}(n, \{\pm 1, n/2\})$, the distance from 0 to j is

$$d(0, j) = \min\{\|j\|_n, 1 + \|j - n/2\|_n\}, \quad (25)$$

where

$$\|t\|_n = \min\{t \bmod n, n - (t \bmod n)\}$$

is cyclic distance on C_n . For $0 \leq j < 4\rho$, (25) gives the following values:

$$d(0, j) = \begin{cases} j, & 0 \leq j \leq \rho, \\ 2\rho + 1 - j, & \rho < j \leq 2\rho, \\ j - 2\rho + 1, & 2\rho < j < 3\rho, \\ 4\rho - j, & 3\rho \leq j < 4\rho. \end{cases}$$

Translation-invariance gives $d(1, j) = d(0, j - 1)$ and $d(n/2, j) = d(0, j - 2\rho)$. Comparing the displayed values yields

$$E_{0,1} = \{-\rho, \rho + 1\}, \quad E_{0,n/2} = \{-\rho, \rho\}.$$

Thus $\eta(0, 1) = \eta(0, n/2) = 2$, and also $\eta(0, -1) = 2$. By the Cayley graph criterion, the graph is NDB.

Conversely, suppose $\gcd(n, a) \neq 1$. Since Γ is connected, the connection set $\{\pm a, n/2\}$ generates $\mathbb{Z}_n$. Equivalently,

$$\gcd(n, a, n/2) = 1.$$

Because $n/2$ divides n , this is the same as

$$\gcd(a, n/2) = 1.$$

Now write $n = 2m$. Since a is relatively prime to $m = n/2$, the number $\gcd(n, a) = \gcd(2m, a)$ is either 1 or 2. We are in the case $\gcd(n, a) \neq 1$, so $\gcd(n, a) = 2$. Hence a is even; combined with $\gcd(a, m) = 1$, this forces $m = n/2$ to be odd, so $n \equiv 2 \pmod{4}$. Moreover, the additive order of a in $\mathbb{Z}_n$ is $n/\gcd(n, a) = n/2 = m$.

Writing $a = 2b$ with $\gcd(b, m) = 1$, the edges generated by $\pm a$ form two m -cycles, one on each parity class, and the generator $m = n/2$ gives a perfect matching between the two cycles. More explicitly, choose $c \in \mathbb{Z}_m$ with $bc \equiv 1 \pmod{m}$ and map

$$2r \mapsto (cr, 0), \quad 2r + m \mapsto (cr, 1)$$

from $\mathbb{Z}_n$ to $\mathbb{Z}_m \times \{0, 1\}$. Under this map the $\pm a$ -edges become the cycle edges $(t, i) \sim (t \pm 1, i)$ and the m -edges become vertical edges $(t, 0) \sim (t, 1)$. Hence $\Gamma \cong C_m \square K_2 = GP(m, 1)$, where m is odd. By the odd-prism example, it is not NDB. $\square$

7. Product constructions for vertex-transitive NDB graphs

The Cartesian product $G \square H$ has vertex set $V(G) \times V(H)$, with (g, h) adjacent to (g', h') if either $g = g'$ and $hh' \in E(H)$, or $h = h'$ and $gg' \in E(G)$. Its distance formula is

$$d_{G \square H}((g, h), (g', h')) = d_G(g, g') + d_H(h, h'). \quad (26)$$

Kutnar and Miklavič characterized NDB Cartesian products in [5, Theorem 4.1]. The next result is the equidistance-profile form for vertex-transitive factors.

Proposition 7.1 (Cartesian products). *Let G and H be finite connected vertex-transitive graphs, both with at least one edge. For an edge $e = gg'$ of G , write*

$$\eta_G(e) = |\{x \in V(G) : d_G(x, g) = d_G(x, g')\}|,$$

and define $\eta_H(f)$ analogously. The equidistance numbers of $G \square H$ are precisely

$$|V(H)|\eta_G(e) \quad (e \in E(G)) \quad \text{and} \quad |V(G)|\eta_H(f) \quad (f \in E(H)).$$

Consequently, $G \square H$ is NDB if and only if G and H are NDB and

$$\frac{\gamma_G}{|V(G)|} = \frac{\gamma_H}{|V(H)|}. \quad (27)$$

In that case

$$\gamma_{G \square H} = |V(H)|\gamma_G = |V(G)|\gamma_H.$$

Proof. Consider an edge $(g, h)(g', h)$ of $G \square H$ coming from an edge gg' of G . By (26), a vertex (x, y) is equidistant from (g, h) and (g', h) if and only if

$$d_G(x, g) + d_H(y, h) = d_G(x, g') + d_H(y, h),$$

that is, if and only if $d_G(x, g) = d_G(x, g')$. Hence the corresponding equidistance number is $|V(H)|\eta_G(gg')$. The argument for edges coming from H gives $|V(G)|\eta_H(f)$.

Since the Cartesian product of vertex-transitive graphs is vertex-transitive, Theorem 5.2 applies. The displayed formula shows that $G \square H$ is NDB if and only if the numbers $|V(H)|\eta_G(e)$ and $|V(G)|\eta_H(f)$ are independent of e and f and are equal. This first forces G and H to be NDB. If G and H are NDB, then

$$\eta_G = |V(G)| - 2\gamma_G, \quad \eta_H = |V(H)| - 2\gamma_H.$$

The equality $|V(H)|\eta_G = |V(G)|\eta_H$ is equivalent to (27), and the formula for $\gamma_{G \square H}$ follows. $\square$

For integers $r, m \geq 1$, let $K_r[m]$ denote the complete multipartite graph with r parts, each of size m .

Lemma 7.2. *For $r \geq 2$ and $m \geq 1$, the graph $K_r[m]$ is vertex-transitive, edge-transitive and NDB. It has*

$$|V(K_r[m])| = rm, \quad \eta = (r - 2)m, \quad \gamma = m.$$

Proof. The automorphism group permutes the r parts and acts symmetrically inside each part, so it is transitive on vertices and on edges. Let uv be an edge, with u and v in two distinct parts. A vertex in one of the remaining

$r - 2$ parts is adjacent to both u and v , and is equidistant from them. A vertex in the part of u is not equidistant from u and v ; the same is true for a vertex in the part of v . Thus $\eta = (r - 2)m$. Theorem 5.2 gives

$$\gamma = \frac{rm - (r - 2)m}{2} = m.$$

□

Proposition 7.3. *Let $r \geq 3$ and let $m, m' \geq 1$ with $m \neq m'$. Then*

$$K_r[m] \square K_r[m']$$

is a non-bipartite vertex-transitive NDB graph which is not edge-transitive. Moreover,

$$\gamma_{K_r[m] \square K_r[m']} = rmm'.$$

Proof. By the preceding lemma,

$$\frac{\gamma_{K_r[m]}}{|V(K_r[m])|} = \frac{m}{rm} = \frac{1}{r} = \frac{m'}{rm'} = \frac{\gamma_{K_r[m']}}{|V(K_r[m'])|}.$$

Proposition 7.1 implies that the Cartesian product is NDB and that its NDB parameter is $rm' \cdot m = rmm'$. It is vertex-transitive because both factors are vertex-transitive, and it is non-bipartite because $r \geq 3$.

It remains to show that the product is not edge-transitive. In a Cartesian product, every triangle lies in one Cartesian direction. An edge of $K_r[m]$ lies in exactly $(r - 2)m$ triangles, while an edge of $K_r[m']$ lies in exactly $(r - 2)m'$ triangles. Since $m \neq m'$, edges in the two Cartesian directions lie in different numbers of triangles and cannot belong to the same automorphism orbit. □

We also record an independent-blow-up construction. Let $\overline{K}_t$ denote the edgeless graph on t vertices. For a graph Γ and an integer $t \geq 1$, the lexicographic product $\Gamma[\overline{K}_t]$ replaces each vertex of Γ by an independent set of size t and each edge by a complete bipartite graph between the corresponding parts. The following result is the special empty-factor case of [5, Theorem 4.4], included because it preserves the vertex-transitive examples above.

Proposition 7.4 (Independent blow-ups). *Let Γ be a connected graph with at least one edge, and let $t \geq 1$. Then $\Gamma[\overline{K}_t]$ is NDB if and only if Γ is NDB. In that case*

$$\gamma_{\Gamma[\overline{K}_t]} = t\gamma_{\Gamma}.$$

If Γ is vertex-transitive, then $\Gamma[\overline{K}_t]$ is vertex-transitive.

Proof. Let $\Lambda = \Gamma[\overline{K_t}]$. Write vertices of Λ as (g, i) with $g \in V(\Gamma)$ and $1 \leq i \leq t$. Every edge of Λ has the form $(u, i)(v, j)$ with $uv \in E(\Gamma)$. For $z \notin \{u, v\}$, all t vertices over z are closer to (u, i) than to (v, j) exactly when $z \in W_{u,v}^\Gamma$. In the fiber over u , only (u, i) is closer to (u, i) than to (v, j) , whereas in the fiber over v , the $t - 1$ vertices other than (v, j) are closer to (u, i) than to (v, j) . These two boundary contributions add to t , which is the contribution of the vertex u to $t|W_{u,v}^\Gamma|$. Therefore

$$|W_{(u,i),(v,j)}^\Lambda| = t|W_{u,v}^\Gamma|.$$

Similarly,

$$|W_{(v,j),(u,i)}^\Lambda| = t|W_{v,u}^\Gamma|.$$

The NDB equivalence and the formula for γ follow. Vertex-transitivity is preserved because automorphisms of Γ and permutations inside the fibers act transitively on the vertices of the blow-up when Γ is vertex-transitive. $\square$

Corollary 7.5. *For $r \geq 3$, $m \neq m'$ and $t \geq 1$, the graph*

$$(K_r[m] \square K_r[m'])[\overline{K_t}]$$

is a non-bipartite vertex-transitive NDB graph. Its NDB parameter is $trmm'$. For every $t \geq 1$ it is not edge-transitive.

Proof. The NDB and vertex-transitivity statements follow from the preceding proposition and the Cartesian-product construction. Non-bipartiteness is inherited from the base graph. The number of common neighbours of an edge in the blow-up is t times the number of common neighbours of the corresponding base edge. Since the two Cartesian directions in the base have different triangle counts, the same remains true after blow-up. Hence the blow-up is not edge-transitive. $\square$

8. The non-SDB side

Kutnar and Miklavíř asked for a non-bipartite NDB graph which is not SDB [5, Problem 3.3]. This question is necessarily outside the vertex-transitive setting, because Lemma 5.1 shows that every finite connected vertex-transitive graph is SDB. The problem was solved by Fernández and Hujdurović [6], who constructed infinite families of non-bipartite NDB graphs that are not SDB.

One of their examples has vertex set

$$\{0, 1, 2\} \times \mathbb{Z}_{10}.$$

For every $j \in \mathbb{Z}_{10}$, its edges are

$$\begin{aligned} (0, j) &\sim (1, j+1), & (0, j) &\sim (1, j+4), \\ (0, j) &\sim (2, j+1), & (0, j) &\sim (2, j+4), \\ (1, j) &\sim (1, j+4), & (2, j) &\sim (2, j+4), \end{aligned}$$

with second coordinates modulo 10. It is a 30-vertex 4-regular non-bipartite NDB graph with $\gamma = 12$, but it is not SDB [6, Proposition 3.7]. Thus the non-SDB problem and the vertex-transitive NDB problem have different metric obstructions: the former requires breaking strong layer-by-layer balance, whereas the latter can only involve variation of the equidistance coordinate over edge orbits.

9. Concluding remarks

The results above give two complementary metric reductions. The direct-product part shows that quasi- λ -balance is governed by the synchronization of even and odd walks in the factors. The vertex-transitive part shows that, once symmetry supplies ordinary balance, the NDB condition is governed by the equidistance coordinate of the edge profile.

Several natural problems remain. The complete bipartite factor theorem suggests studying the arc invariants α_H and β_H for larger families of non-bipartite graphs, especially distance-regular and arc-transitive graphs. The Cayley criterion asks for a structural understanding of inverse-closed generating sets S for which the values

$$|\{g \in G : \ell_S(g) = \ell_S(s^{-1}g)\}|$$

are independent of $s \in S$. Finally, the product constructions in Section 7 show that non-edge-transitive vertex-transitive NDB graphs are abundant; a classification by edge-orbit data, even for restricted Cayley families, would sharpen the vertex-transitive problem further.

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